



University of Bahrain

Information Technology College

Information System Department

ITIS213 Section No.03

Travel Agency



Group No.03

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Introduction

This project seeks to achieve smooth functioning of the processes involved in this system, with our new database management system. We aim to satisfy both our customers as well as ourselves as a company. Since our institution is a travel agency, multiple obstacles and difficulties must be taken into consideration, which evolves around the variety of countries, airlines, packages, booking, reservations, and batches of people. All this requires a careful and detailed management system.

Background

Years ago, our friend group decided to travel to the United Kingdom during the mid-term break for about a week. We booked with a travel agency to have an organized and fun trip. However, we faced a few problems like name typos, and our luggage was sent to inaccurate destinations, as well as other problems that led us to miss our flights. Because of this, we were forced to book another flight to reach our destination. Therefore, we had multiple brainstorming sessions and produced the idea of developing a travel agency that focuses on avoiding mistakes like these. During our trip to the United Kingdom, we came across a famous travel agency and we thought of franchising that agency in Bahrain ensuring a high quality and smooth route to specific destinations.

Our agency works with all countries around the globe which are accepted by our Ministry of Foreign Affairs, as well as one specific airline: Gulf Air. Our services focus on quick response to our customers and increasing the number of employees in the call center to avoid long hold periods. We also care about the accuracy of data in our systems which contributes to limiting silly mistakes that later become responsible for huge problems. These include inaccurate names, passport numbers, dates, and times. In addition, a vital element is security and the advancement of our technology that offers innovative reporting tools and customizable dashboards. These machineries also require well-trained staff that can diligently deal with these devices. All this contributes to properly organizing all aspects of the travel agency database management system.

Problem Description

For a travel agency, we can adapt the concept to fit the travel industry. We can have multiple office locations or franchises. Each office can offer travel packages and handle a certain number of clients. Travel agents can assist a maximum number of clients with their travel plans. The travel agency can offer different travel packages such as silver, gold, or premium travel. Each client can choose a travel package that suits their preferences. The key is to adapt the concept of managing clients, services, and payments to fit the travel agency model while ensuring that the system can handle the specific requirements of the travel industry.

User Requirements

Our database system offers a variety of packages for diverse groups of people. For instance, people can choose to join the silver membership, which comes with economy-class flights and 4-star hotels. As for the gold package, the characteristics that come with it are business-class flights and 4-star hotels that consist of special features like a gym, outdoor pool, and evening entertainment activities. Lastly, the premium membership incorporates first-class flights with special menus and massive spaces, as well as 5-star hotels that include high security, a private pool, a spa, and an incredible view.

To ensure that our customers have a safe and easy flight free of obstacles, we offer health insurance, visas, as well as economy, business, and first-class flight reservations.

Data Element

Table: Employee

(STAFF_ID, FNAME, LNAME, EMAIL, PHONE_NUMBER, BRANCH_ID)

Table: Customer

(CUSTOMER_ID, FNAME, LNAME, EMAIL, PHONE, PASSPORT, PACKAGE_ID)

Table: Booking

(BOOKING_ID, CUSTOMER_ID, PACKAGE_ID, ARRIVAL_DATE, DEPARTURE_DATE, HOTEL_ID, FLIGHT_ID, TOTAL_COST)

Table: Flight

(FLIGHT_ID, AIRLINE, COUNTRY, CLASS_TYPE, PACKAGE_ID, PRICE)

Table: Hotel

(HOTEL_ID, HOTEL_NAME, LOCATION, PACKAGE_ID, NUM_OF_STAR, ROOM_TYPE, PRICE_PER_NIGHT, COUNTRY)

Table: CarRental

(BOOKING_ID, CAR_COMPANY, PICKUP_DATE, RETURN_DATE, NUM_OF_DAY, PRICE_PER_DAY)

Table: Package

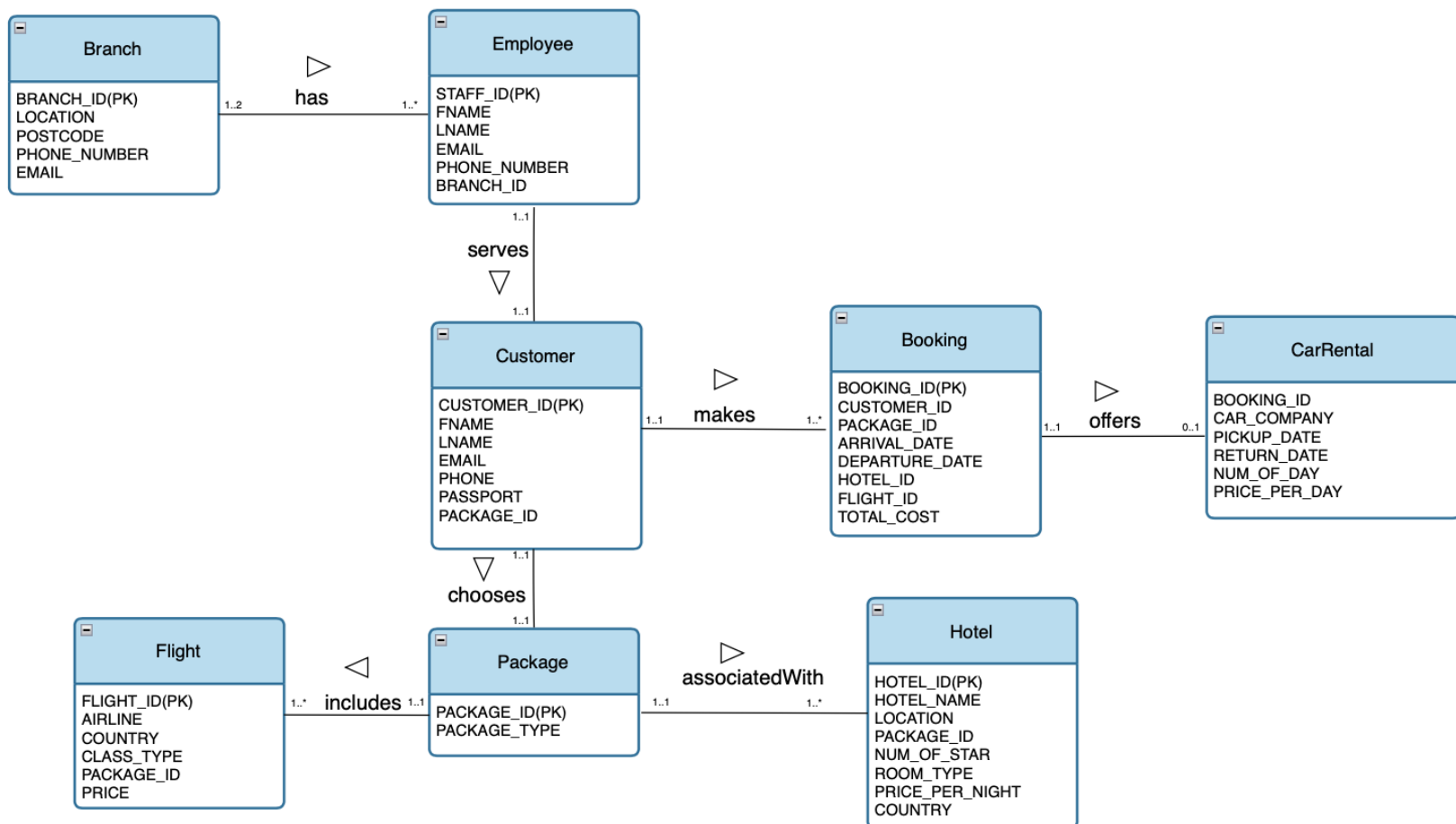
(PACKAGE_ID, PACKAGE_TYPE)

Table: Branch

(BRANCH_ID, LOCATION, POSTCODE, PHONE_NUMBER, EMAIL)

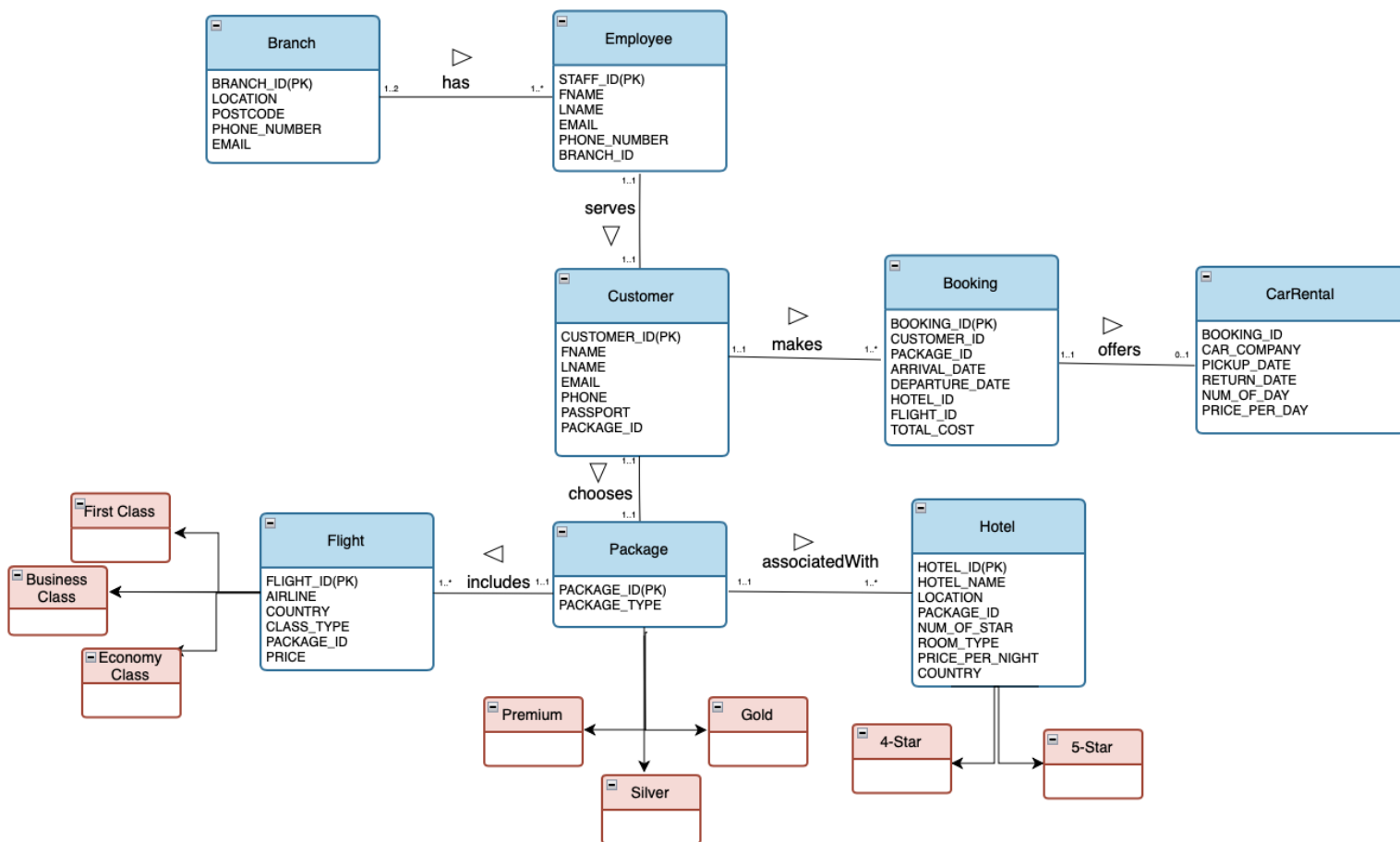
ER Diagram:

Our travel agency consists of 2 branches. Each of these branches has one to many employees, where every employee serves exactly one customer. Customers can make several bookings, which also offers optional car rental services. In addition, customers can choose a package that suits their preferences. The packages available include flights and are associated with hotels.



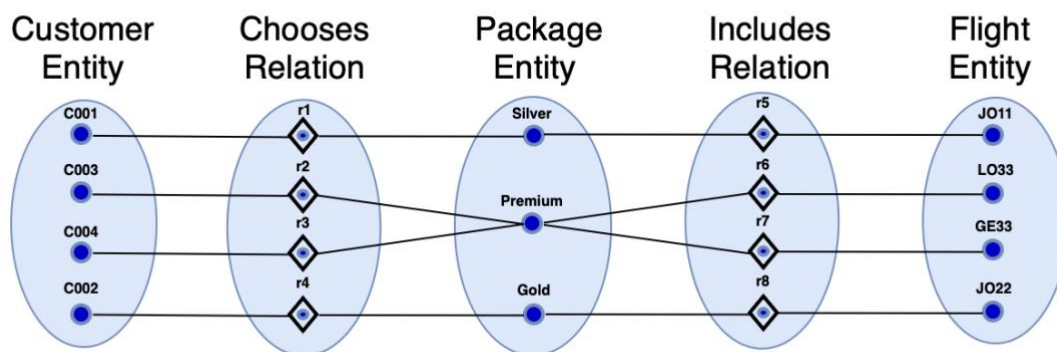
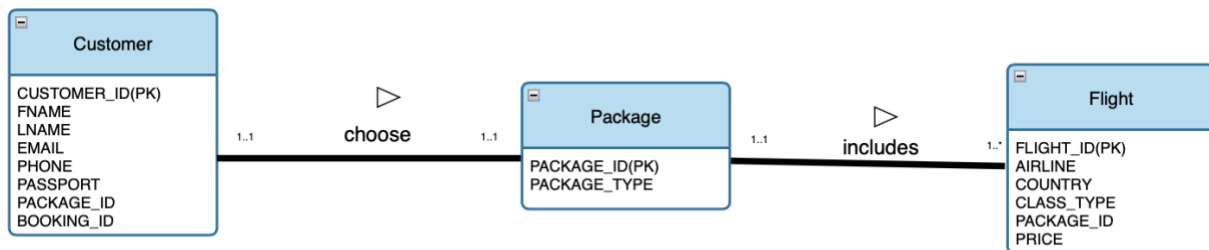
EER Diagram:

Our flights are categorized into three classes, which are first-class, business-class, and economy-class. Our packages also include three different packages: silver, gold, and premium. Lastly, the hotels we offer are either 4-star or 5-star hotels.

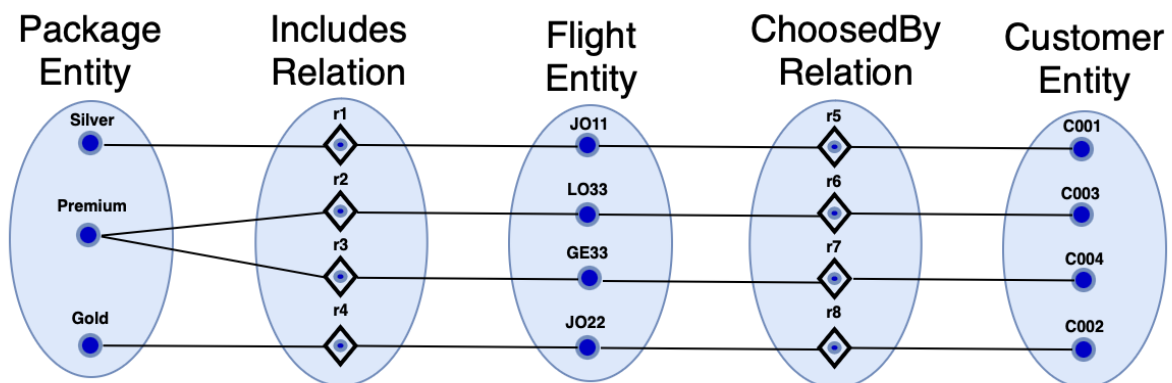
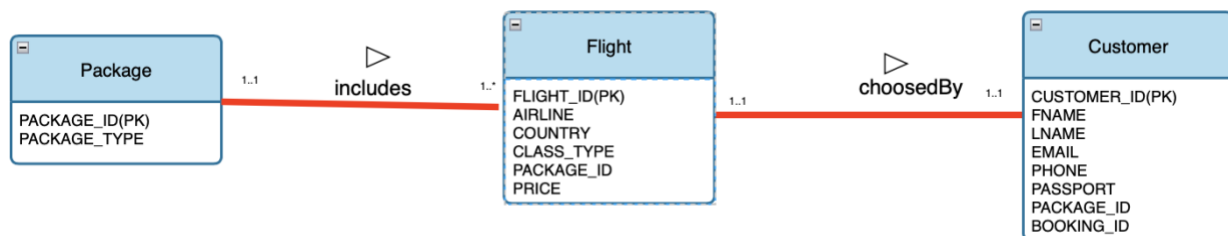


Connection Traps:

The trap displayed below is a fan trap. A fan trap occurs when the path between the entities may seem unclear, causing inaccurate relationships. In our trap, it is not clear which customer has which flight since the pathways intersect.



This problem was solved by rearranging the entities, where the customer entity and the relationship chosen were moved to the other side and a few modifications were made to create a complete and solved semantic net.



Normalization:

In the normalization process, we have three levels: 1NF, 2NF, and 3NF.

The first normalized form (1NF) involves removing any repeating groups available in the tables.

The second normalized form (2NF) requires removing the partial functional dependencies (FD).

Lastly, the third normalized form (3NF) includes removing the transitive FDs.

We have one unnormalized table which is the Booking table:

booking_id	customer_id	package_id	arrival_date	departure_date	hotel_id	flight_id	total_cost
B001	C001	A100	12/26/2023	12/31/2023	GER1	GE11	1350
B002					JOR1	JO11	1120
B003	C002	B200	12/27/2023	12/30/2023	JOR2	JO22	1360
B004	C003	C300	12/28/2023	12/31/2023	LON3	LO33	2100
B005	C004		12/29/2023		GER3	GE33	2260

As displayed above, we have several repeating groups that need to be removed.

1NF:

booking_id	customer_id	package_id	arrival_date	departure_date	hotel_id	flight_id	total_cost
B001	C001	A100	12/26/2023	12/31/2023	GER1	GE11	1350
B002	C001	A100	12/26/2023	12/31/2023	JOR1	JO11	1120
B003	C002	B200	12/27/2023	12/30/2023	JOR2	JO22	1360
B004	C003	C300	12/28/2023	12/31/2023	LON3	LO33	2100
B005	C004	C300	12/29/2023	12/31/2023	GER3	GE33	2260

The table is now free from any repeating groups and is normalized to the first form.

The next step is to eliminate the partial FDs.

2NF:

These are the partial FDs:

Branch table:

branch_id $\longrightarrow$ location, postcode, phone_number, email (all attributes)

Employee table:

staff_id $\longrightarrow$ fname, lname, email, phone_number, branch_id (all attributes)

Customer table:

Customer_id $\longrightarrow$ fname, lname, email, phone, passport, package_id (all attributes)

Booking table:

booking_id $\longrightarrow$ arrival_date, departure_date, hotel_id, flight_id, total_cost

customer_id $\longrightarrow$ package_id

Package table:

package_id $\longrightarrow$ package_type (all attributes)

Flight table:

flight_id $\longrightarrow$ airline, country

package_id $\longrightarrow$ class_type

flight_id, package_id $\longrightarrow$ price

Hotel table:

hotel_id → hotel_name, location, room_type, country

package_id → num_of_star

hotel_id, package_id → price_per_night

Car Rental table:

booking_id → car_company, pickup_date, return_date, num_of_day, price_per_day (all attributes)

The tables branch, employee, customer, package, and car rental have no changes since all attributes were dependent on the primary key.

New tables created from the booking table:

Booking table:

booking_id	arrival_date	departure_date	hotel_id	flight_id	total_cost
B001	12/26/2023	12/31/2023	GER1	GE11	1350
B002	12/26/2023	12/31/2023	JOR1	JO11	1120
B003	12/27/2023	12/30/2023	JOR2	JO22	1360
B004	12/28/2023	12/31/2023	LON3	LO33	2100
B005	12/29/2023	12/31/2023	GER3	GE33	2260

Package table:

customer_id	package_id
C001	A100
C001	A100
C002	B200
C003	C300
C004	C300

New tables created from the flight table:

Flight table:

flight_id	airline	country
LO11	Gulf Air	London
LO22	Gulf Air	London
LO33	Gulf Air	London
NY11	Gulf Air	New York
NY22	Gulf Air	New York
NY33	Gulf Air	New York
PA11	Gulf Air	Paris
PA22	Gulf Air	Paris
PA33	Gulf Air	Paris
GE11	Gulf Air	Germany
GE22	Gulf Air	Germany
GE33	Gulf Air	Germany
JO11	Gulf Air	Jordan
JO22	Gulf Air	Jordan
JO33	Gulf Air	Jordan

Package table:

package_id	class_type
A100	Economy
B200	Business
C300	First Class
A100	Economy
B200	Business
C300	First Class
A100	Economy
B200	Business
C300	First Class
A100	Economy
B200	Business
C300	First Class
A100	Economy
B200	Business
C300	First Class

Price table:

flight_id	package_id	price
LO11	A100	500
LO22	B200	800
LO33	C300	1200
NY11	A100	600
NY22	B200	900
NY33	C300	1300
PA11	A100	550
PA22	B200	850
PA33	C300	1250
GE11	A100	700
GE22	B200	1000
GE33	C300	1500
JO11	A100	520
JO22	B200	820
JO33	C300	1220

New tables created from hotel table:

Hotel table:

hotel_id	hotel_name	location	room_type	country
LON1	Four Season	London	Standard	London
LON2	The Ritz	London	Deluxe	London
LON3	Zedwell	London	Suite	London
NY10	Hayyat Palace	56GF 56 GH67	Standard	New York
NY20	Delta	HY76 98 VF00	Deluxe	New York
NY30	Pod51	BK67 45 65TG	Suite	New York
PAR2	Novotel	6795 98H	Deluxe	Paris
GER1	Liebesbier	GT68 87	Standard	Germany
GER2	Four Side	FM78 33	Deluxe	Germany
GER3	Prizeotel	HQ56 93	Suite	Germany
JOR1	Crown Plaza	TV67094	Standard	Jordan
JOR2	Shirat	67MB 89 TY80	Deluxe	Jordan
JOR3	Jordan Season	LMQZD8109	Suite	Jordan
PAR1	The Ritz Paris	4567 G7Y	Standard	Paris
PAR3	Four Season	5688 DC9	Suite	Paris

Stars table:

package_id	num_of_star
A100	4
B200	4
C300	5
A100	4
B200	4
C300	5
B200	4
A100	4
B200	4
C300	5
A100	4
B200	4
C300	5
A100	4
C300	5

Price table:

hotel_id	package_id	price_per_night
LON1	A100	150
LON2	B200	200
LON3	C300	300
NY10	A100	160
NY20	B200	220
NY30	C300	320
PAR2	B200	210
GER1	A100	130
GER2	B200	190
GER3	C300	380
JOR1	A100	120
JOR2	B200	180
JOR3	C300	400
PAR1	A100	140
PAR3	C300	350

3NF:

The tables in 2NF are already in their simplest form and cannot be simplified more, therefore they are already in 3NF.

Creation of Table

create table BRANCH (

"BRANCH_ID" char(4) not null,

"LOCATION" varchar(20),

"POSTCODE" number,

"PHONE_NUMBER" number,

"EMAIL" varchar(50),

constraint "BRANCH_PK" primary key ("BRANCH_ID"));

INSERT INTO BRANCH VALUES ('a123', 'riffa', 1234, 12345678,
TravelAgency213@gmail.com);

INSERT INTO BRANCH VALUES ('b456', 'muharraq', 4567, 98765432,
TravelAgency213@gmail.com);

BRANCH_ID	LOCATION	POSTCODE	PHONE_NUMBER	EMAIL
a123	riffa	1234	12345678	TravelAgency213@gmail.com
b456	muharraq	4567	98765432	TravelAgency213@gmail.com

CREATE TABLE EMPLOYEE (

STAFF_ID CHAR(4) NOT NULL,

FNAME VARCHAR(10),

LNAME VARCHAR(10),

```

EMAIL      VARCHAR(50),

PHONE_NUMBER NUMBER,

BRANCH_ID  CHAR(4) NOT NULL,

CONSTRAINT EMPLOYEE_PK PRIMARY KEY (STAFF_ID),

CONSTRAINT FK_BRANCH_ID FOREIGN KEY (BRANCH_ID) REFERENCES
BRANCH(BRANCH_ID) );

INSERT INTO EMPLOYEE VALUES ('E1', 'John', 'Doe', 'john.doe@example.com', 12345678,
'b456');

INSERT INTO EMPLOYEE VALUES ('E2', 'Jane', 'Smith', 'jane.smith@example.com',
23456789, 'b456');

INSERT INTO EMPLOYEE VALUES ('E3', 'Michael', 'Johnson',
'michael.johnson@example.com', 34567890, 'b456');

INSERT INTO EMPLOYEE VALUES ('E4', 'Emily', 'Brown', 'emily.brown@example.com',
45678901, 'b456');

INSERT INTO EMPLOYEE VALUES ('E5', 'David', 'Lee', 'david.lee@example.com',
56789012, 'b456');

INSERT INTO EMPLOYEE VALUES ('E6', 'Sarah', 'Wilson', 'sarah.wilson@example.com',
67890123, 'a123');

INSERT INTO EMPLOYEE VALUES ('E7', 'Matthew', 'Martinez',
'matthew.martinez@example.com', 78901234, 'a123');

```

```
INSERT INTO EMPLOYEE VALUES ('E8', 'Olivia', 'Garcia', 'olivia.garcia@example.com',
89012345, 'a123');
```

```
INSERT INTO EMPLOYEE VALUES ('E9', 'Daniel', 'Lopez', 'daniel.lopez@example.com',
90123456, 'a123');
```

```
INSERT INTO EMPLOYEE VALUES ('E10', 'Sophia', 'Perez', 'sophia.perez@example.com',
12345678, 'a123');
```

STAFF_ID	FNAME	LNAME	EMAIL	PHONE_NUMBER	BRANCH_ID
E1	John	Doe	john.doe@example.com	12345678	b456
E2	Jane	Smith	jane.smith@example.com	23456789	b456
E3	Michael	Johnson	michael.johnson@example.com	34567890	b456
E4	Emily	Brown	emily.brown@example.com	45678901	b456
E5	David	Lee	david.lee@example.com	56789012	b456
E6	Sarah	Wilson	sarah.wilson@example.com	67890123	a123
E7	Matthew	Martinez	matthew.martinez@example.c...	78901234	a123
E8	Olivia	Garcia	olivia.garcia@example.com	89012345	a123
E9	Daniel	Lopez	daniel.lopez@example.com	90123456	a123
E10	Sophia	Perez	sophia.perez@example.com	12345678	a123

```
create table PACKAGE (
    "PACKAGE_ID" char(4) not null,
    "PACKAGE_TYPE" varchar(10),
    constraint "PACKAGE_PK" primary key ( "PACKAGE_ID" ) );
```

```
INSERT INTO PACKAGE VALUES ('a100', 'silver');
```

```
INSERT INTO PACKAGE VALUES ('b200', 'gold');
```

```
INSERT INTO PACKAGE VALUES ('c300', 'premium');
```

PACKAGE_ID	PACKAGE_TYPE
a100	silver
b200	gold
c300	premium

```

CREATE TABLE HOTEL (

    HOTEL_ID      CHAR(4),

    HOTEL_NAME    VARCHAR(30),

    LOCATION      VARCHAR(50),

    PACKAGE_ID     CHAR(4),

    NUM_OF_STAR   NUMBER,

    ROOM_TYPE     VARCHAR(10),

    PRICE_PER_NIGHT NUMBER,

    COUNTRY VARCHAR(20),

    CONSTRAINT HOTEL_PK PRIMARY KEY (HOTEL_ID),

    CONSTRAINT FK_PACKAGE_ID FOREIGN KEY (PACKAGE_ID) REFERENCES
PACKAGE(PACKAGE_ID) );

INSERT INTO HOTEL VALUES ('LON1', 'Four Season', 'London', 'a100', 4, 'Standard',
150,'London');

```

INSERT INTO HOTEL VALUES('LON2', 'The Ritz', 'London', 'b200', 4, 'Deluxe',
200,'London');

INSERT INTO HOTEL VALUES('LON3', 'Zedwell', 'London', 'c300', 5, 'Suite', 300,'London');

INSERT INTO HOTEL VALUES ('NY10', 'Hayyat Place', '56GF 56 GH67', 'a100', 4, 'Standard',
160, 'New York');

INSERT INTO HOTEL VALUES('NY20', 'Delta', 'HY76 98 VF00', 'b200', 4, 'Deluxe', 220
, 'New York');

INSERT INTO HOTEL VALUES('NY30', 'Pod51', 'BK67 45 65TG', 'c300', 5, 'Suite', 320, 'New
York');

INSERT INTO HOTEL VALUES ('PAR1', 'THE RITZ PARIS', '4567 67Y', 'a100', 4, 'Standard',
140, 'Paris');

INSERT INTO HOTEL VALUES('PAR2', 'Novotel', '6795 98H', 'b200', 4, 'Deluxe', 210,'Paris');

INSERT INTO HOTEL VALUES('PAR3', 'Four Season', '5688 DC9', 'c300', 5, 'Suite',
350m'Paris');

INSERT INTO HOTEL VALUES ('GER1', 'Liebesbier', 'GT68 87', 'a100', 4, 'Standard',
130,'Germany');

```
INSERT INTO HOTEL VALUES('GER2', 'FourSIDE', 'FM78 33', 'b200', 4, 'Deluxe',
190,'Germany');
```

```
INSERT INTO HOTEL VALUES('GER3', 'Prizeotel', 'HQ56 93', 'c300', 5, 'Suite',
380,'Germany');
```

```
INSERT INTO HOTEL VALUES ('JOR1', 'Crown Plaza', 'TV67094', 'a100', 4, 'Standard',
120,'Jordan');
```

```
INSERT INTO HOTEL VALUES ('JOR2', 'Shirat', '', 'b200', 4, 'Deluxe', 180,'Jordan');
```

```
INSERT INTO HOTEL VALUES ('JOR3', 'Jordan Season', 'LMQZD8109', 'c300', 5, 'Suite',
400,'Jordan');
```

HOTEL_ID	HOTEL_NAME	LOCATION	PACKAGE_ID	NUM_OF_STAY	ROOM_TYPE	PRICE_PER_NIGHT	COUNTRY
LON1	Four Season	London	a100	4	Standard	150	London
LON2	The Ritz	London	b200	4	Deluxe	200	London
LON3	Zedwell	London	c300	5	Suite	300	London
NY10	Hayes Place	56GF 56 GH07	a100	4	Standard	160	New York
NY20	Delta	HY76 WE VF00	b200	4	Deluxe	220	New York
NY30	Pud51	8K07 45 65TG	c300	5	Suite	320	New York
PAR2	Navotel	6795 98H	b200	4	Deluxe	210	Paris
GER1	Liebschaler	GT68 87	a100	4	Standard	130	Germany
GER2	FourSIDE	FM78 33	b200	4	Deluxe	190	Germany
GER3	Prizeotel	HQ56 93	c300	5	Suite	380	Germany
JOR1	Crown Plaza	TV67094	a100	4	Standard	120	Jordan
JOR2	Shirat		b200	4	Deluxe	180	Jordan
JOR3	Jordan Season	LMQZD8109	c300	5	Suite	400	Jordan
PAR1	THE RITZ PARIS	4567 67Y	a100	4	Standard	140	Paris
PAR3	Four Season	5688 DC9	c300	5	Suite	350	Paris

```
CREATE TABLE FLIGHT (
```

```
    FLIGHT_ID CHAR(4) NOT NULL,
```

```
    PACKAGE_ID CHAR(4) NOT NULL,
```

```
    AIRLINE VARCHAR2(20),
```

```
    COUNTRY VARCHAR2(30),
```

```
    CLASS_TYPE VARCHAR2(30),
```

```

PRICE    NUMBER,

CONSTRAINT FLIGHT_PK PRIMARY KEY (FLIGHT_ID),

CONSTRAINT FK2_PACKAGE_ID FOREIGN KEY (PACKAGE_ID) REFERENCES
PACKAGE(PACKAGE_ID)

);

INSERT INTO VALUES ('LO11', 'a100', 'Gulfair', 'LONDON', 'Economy', 500);

INSERT INTO FLIGHT VALUES ('LO22', 'b200', 'Gulfair', 'LONDON', 'Business', 800);

INSERT INTO FLIGHT VALUES ('LO33', 'c300', 'Gulfair', 'LONDON', 'First Class', 1200);

INSERT INTO FLIGHT VALUES ('NY11', 'a100', 'Gulfair', 'NEW YORK', 'Economy', 600);

INSERT INTO FLIGHT VALUES ('NY22', 'b200', 'Gulfair', 'NEW YORK', 'Business', 900);

INSERT INTO FLIGHT VALUES ('NY33', 'c300', 'Gulfair', 'NEW YORK', 'First Class', 1300);

INSERT INTO FLIGHT VALUES ('PA11', 'a100', 'Gulfair', 'PARIS', 'Economy', 550);

INSERT INTO FLIGHT VALUES ('PA22', 'b200', 'Gulfair', 'PARIS', 'Business', 850);

INSERT INTO FLIGHT VALUES ('PA33', 'c300', 'Gulfair', 'PARIS', 'First Class', 1250);

INSERT INTO FLIGHT VALUES ('GE11', 'a100', 'Gulfair', 'GERMANY', 'Economy', 700);

INSERT INTO FLIGHT VALUES ('GE22', 'b200', 'Gulfair', 'GERMANY', 'Business', 1000);

INSERT INTO FLIGHT VALUES ('GE33', 'c300', 'Gulfair', 'GERMANY', 'First Class', 1500);

INSERT INTO FLIGHT VALUES ('JO11', 'a100', 'Gulfair', 'JORDAN', 'Economy', 520);

```

INSERT INTO FLIGHT VALUES ('JO22', 'b200', 'Gulfair', 'JORDAN', 'Business', 820);

INSERT INTO FLIGHT VALUES ('JO33', 'c300', 'Gulfair', 'JORDAN', 'First Class', 1220);

FLIGHT_ID	PACKAGE_ID	AIRLINE	COUNTRY	CLASS_TYPE	PRICE
LO11	a100	Gulfair	LONDON	Economy	500
LO22	b200	Gulfair	LONDON	Business	800
LO33	c300	Gulfair	LONDON	First Class	1200
NY11	a100	Gulfair	NEW YORK	Economy	600
NY22	b200	Gulfair	NEW YORK	Business	900
NY33	c300	Gulfair	NEW YORK	First Class	1800
PA11	a100	Gulfair	PARIS	Economy	550
PA22	b200	Gulfair	PARIS	Business	850
PA33	c300	Gulfair	PARIS	First Class	1250
GE11	a100	Gulfair	GERMANY	Economy	700
GE22	b200	Gulfair	GERMANY	Business	1000
GE33	c300	Gulfair	GERMANY	First Class	1900
JO11	a100	Gulfair	JORDAN	Economy	520
JO22	b200	Gulfair	JORDAN	Business	820
JO33	c300	Gulfair	JORDAN	First Class	1220

CREATE TABLE CUSTOMER (

CUSTOMER_ID CHAR(4) NOT NULL,

FNAME VARCHAR(20),

LNAME VARCHAR(20),

EMAIL VARCHAR(50),

PHONE NUMBER(15),

PASSPORT NUMBER(10),

PACKAGE_ID CHAR(4),

CONSTRAINT CUSTOMER_PK PRIMARY KEY (CUSTOMER_ID),

CONSTRAINT FK_PACKAGE_ID3 FOREIGN KEY (PACKAGE_ID) REFERENCES

PACKAGE(PACKAGE_ID)

);

```
INSERT INTO CUSTOMER VALUES ('C001', 'John', 'Doe', 'john.doe@example.com',  
1234567890, 1234567890, 'a100');
```

```
INSERT INTO CUSTOMER VALUES ('C002', 'Jane', 'Smith', 'jane.smith@example.com',  
2345678901, 2345678901, 'b200');
```

```
INSERT INTO CUSTOMER VALUES('C003', 'Michael', 'Johnson',  
'michael.johnson@example.com', 3456789012, 3456789012, 'c300');
```

```
INSERT INTO CUSTOMER VALUES('C004', 'Emily', 'Davis', 'emily.davis@example.com',  
4567890123, 4567890123, 'c300');
```

CUSTOMER_ID	FNAME	LNAME	EMAIL	PHONE	PASSPORT	PACKAGE_ID
C001	John	Doe	john.doe@example.com	1234567890	1234567890	a100
C002	Jane	Smith	jane.smith@example.com	2345678901	2345678901	b200
C003	Michael	Johnson	michael.johnson@exampl...	3456789012	3456789012	c300
C004	Emily	Davis	emily.davis@example.com	4567890123	4567890123	c300

```
CREATE TABLE BOOKING (
```

```
    BOOKING_ID    CHAR(4) NOT NULL,
```

```
    CUSTOMER_ID   CHAR(4),
```

```
    PACKAGE_ID    CHAR(4),
```

```
    ARRIVAL_DATE  DATE,
```

```
    DEPARTURE_DATE DATE,
```

```
    HOTEL_ID      CHAR(4),
```

```

FLIGHT_ID    CHAR(4),

TOTAL_COST    NUMBER,

CONSTRAINT BOOKING_PK PRIMARY KEY (BOOKING_ID),

CONSTRAINT FK_CUSTOMER_ID2 FOREIGN KEY (CUSTOMER_ID) REFERENCES
CUSTOMER(CUSTOMER_ID),

CONSTRAINT FK_PACKAGE_ID2 FOREIGN KEY (PACKAGE_ID) REFERENCES
PACKAGE(PACKAGE_ID),

CONSTRAINT FK_HOTEL_ID2 FOREIGN KEY (HOTEL_ID) REFERENCES
HOTEL(HOTEL_ID),

CONSTRAINT FK_FLIGHT_ID2 FOREIGN KEY (FLIGHT_ID) REFERENCES
FLIGHT(FLIGHT_ID)

);

INSERT INTO BOOKING VALUES ('B001', 'C001', 'a100', TO_DATE('2023-12-26','YYYY-
MM-DD'), TO_DATE('2023-12-31', 'YYYY-MM-DD'), 'GER1', 'GE11',1350),

INSERT INTO BOOKING VALUES ('B002', 'C001', 'a100', TO_DATE('2023-12-26',
'YYYY-MM-DD'), TO_DATE('2023-12-31', 'YYYY-MM-DD'), 'JOR1', 'JO11',1120),

INSERT INTO BOOKING VALUES ('B003', 'C002', 'b200', TO_DATE('2023-12-27',
'YYYY-MM-DD'), TO_DATE('2023-12-30', 'YYYY-MM-DD'), 'JOR2', 'JO22',1360),

INSERT INTO BOOKING VALUES('B004', 'C003', 'c300', TO_DATE('2023-12-28','YYYY-
MM-DD'), TO_DATE('2023-12-31','YYYY-MM-DD'),'LON3','LO33',2100);

```

```
INSERT INTO BOOKING VALUES ('B005', 'C004', 'c300', TO_DATE('2023-12-29',
'YYYY-MM-DD'), TO_DATE('2023-12-31', 'YYYY-MM-DD'), 'GER3', 'GE33', 2260);
```

BOOKING_ID	CUSTOMER_ID	PACKAGE_ID	ARRIVAL_DATE	DEPARTURE_DATE	HOTEL_ID	FLIGHT_ID	TOTAL_COST
B002	C001	a100	12/26/2023	12/31/2023	JOR1	J011	1120
B003	C002	b200	12/27/2023	12/30/2023	JOR2	J022	1360
B005	C004	c300	12/29/2023	12/31/2023	GER3	GE33	2260
B004	C003	c300	12/28/2023	12/31/2023	LON3	L033	2100
B001	C001	a100	12/26/2023	12/31/2023	GER1	GE11	1350

```
CREATE TABLE CARRENTAL (

    BOOKING_ID    CHAR(4) NOT NULL,

    CAR_COMPANY    VARCHAR(20),

    PICKUP_DATE    DATE,

    RETURN_DATE    DATE,

    NUM_OF_DAY    NUMBER,

    PRICE_PER_DAY    NUMBER,

    CONSTRAINT FK_BOOKING_ID3 FOREIGN KEY (BOOKING_ID) REFERENCES
BOOKING(BOOKING_ID)

);
```

```
INSERT INTO CARRENTAL VALUES ('B003', 'Lexus', TO_DATE('2023-12-27', 'YYYY-
MM-DD'), TO_DATE('2023-12-30', 'YYYY-MM-DD'), 3, 50);
```

```
INSERT INTO CARRENTAL VALUES ('B005', 'Mercedes', TO_DATE('2023-12-29', 'YYYY-
MM-DD'), TO_DATE('2023-12-31', 'YYYY-MM-DD'), 2, 60);
```

BOOKING_ID	CAR_COMPANY	PICKUP_DATE	RETURN_DATE	NUM_OF_DAY	PRICE_PER_DAY
B003	Lexus	12/27/2023	12/30/2023	3	50
B005	Mercedes	12/29/2023	12/31/2023	2	60

SQL Queries

Create at least 20 different queries (The queries you may develop, should show the following: A natural language description of the query, The SQL commands, and the result generated by SQL. The 20 queries should include: Multiple tables join; Use sub-queries; Group data using GROUP BY; Sort query results using ORDER BY; IN or NOT IN; LIKE; Aggregate functions or calculated fields.

1) Write a query uses a join between the CUSTOMER and BOOKING tables to identify customers that choosed the gold package.

```
SELECT C.FNAME, C.LNAME, C.PACKAGE_ID
FROM CUSTOMER C
INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID
INNER JOIN PACKAGE P ON B.PACKAGE_ID = P.PACKAGE_ID
WHERE P.PACKAGE_TYPE = 'gold';
```

FNAME	LNAME	PACKAGE_ID
Jane	Smith	b200

2) Write query filters the FLIGHT table based on the airline (Gulfair) and price (greater than \$1000) to identify corresponding flights.

```
SELECT FLIGHT_ID, PACKAGE_ID, AIRLINE, COUNTRY, CLASS_TYPE, PRICE
FROM FLIGHT
WHERE AIRLINE = 'Gulfair' AND PRICE > 1000;
```

FLIGHT_ID	PACKAGE_ID	AIRLINE	COUNTRY	CLASS_TYPE	PRICE
L033	c300	Gulfair	LONDON	First Class	1200
NY33	c300	Gulfair	NEW YORK	First Class	1300
PAR3	c300	Gulfair	PARIS	First Class	1250
GE33	c300	Gulfair	GERMANY	First Class	1500
JO33	c300	Gulfair	JORDAN	First Class	1220

3) Write query to produce a list of hotels in descending order of price per night

```
SELECT HOTEL_NAME, ROOM_TYPE, PRICE_PER_NIGHT  
  
FROM HOTEL  
  
ORDER BY PRICE_PER_NIGHT DESC;
```

HOTEL_NAME	ROOM_TYPE	PRICE_PER_NIGHT
Jordan Season	Suite	400
Prizeotel	Suite	380
Four Season	Suite	350
Pod51	Suite	320
Zedwell	Suite	300
Delta	Deluxe	220

4) Write query to display customer first and last names, IDs, and car rental details:

```
SELECT C.FNAME, C.LNAME, C.CUSTOMER_ID, CR. CAR_COMPANY, CR.PICKUP_DATE,  
CR.RETURN_DATE, CR.NUM_OF_DAY, CR.PRICE_PER_DAY  
  
FROM CUSTOMER C  
  
INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID  
  
INNER JOIN CARRENTAL CR ON B.BOOKING_ID = CR.BOOKING_ID;
```

5) calculate the average of the total cost in the booking table

```
SELECT AVG(TOTAL_COST) AS Average_Total_Cost  
  
FROM BOOKING;
```

AVERAGE_TOTAL_COST
1638

6) query to list all customers who booked trips to Jordan:

```
SELECT C.FNAME, C.LNAME, C.CUSTOMER_ID, B.HOTEL_ID, F.FLIGHT_ID  
  
FROM CUSTOMER C
```

INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID

INNER JOIN HOTEL H ON B.HOTEL_ID = H.HOTEL_ID

INNER JOIN FLIGHT F ON B.FLIGHT_ID = F.FLIGHT_ID

WHERE H.COUNTRY = 'Jordan'

OR F.COUNTRY = 'Jordan';

FNAME	LNAME	CUSTOMER_ID	HOTEL_ID	FLIGHT_ID
John	Doe	C001	J0R1	J011
Jane	Smith	C002	J0R2	J022

7) calculate the cost of car rental for each customer:

SELECT C.FNAME, C.LNAME, CR.BOOKING_ID, CR.CAR_COMPANY, CR.PRICE_PER_DAY,
CR.NUM_OF_DAY,

CR.PRICE_PER_DAY * CR.NUM_OF_DAY AS Rental_Cost

FROM CUSTOMER C

INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID

INNER JOIN CARRENTAL CR ON B.BOOKING_ID = CR.BOOKING_ID;

FNAME	LNAME	BOOKING_ID	CAR_COMPANY	PRICE_PER_DAY	NUM_OF_DAY	RENTAL_COST
Jane	Smith	B005	Lexus	50	3	150
Emily	Davis	B005	Mercedes	60	2	120

2 rows returned in 0.01 seconds [Download](#)

8) display the flights included in the Silver package:

SELECT FLIGHT_ID, AIRLINE, COUNTRY, CLASS_TYPE, PRICE

FROM FLIGHT

JOIN PACKAGE ON FLIGHT.PACKAGE_ID = PACKAGE.PACKAGE_ID

WHERE PACKAGE_TYPE = 'silver';

FLIGHT_ID	AIRLINE	COUNTRY	CLASS_TYPE	PRICE
LO11	Gulfair	LONDON	Economy	500
NY11	Gulfair	NEW YORK	Economy	600
PA11	Gulfair	PARIS	Economy	550
GE11	Gulfair	GERMANY	Economy	700
JO11	Gulfair	JORDAN	Economy	520

5 rows returned in 0.01 seconds [Download](#)

9) display the employees working in the A123 branch, along with the branch location:

SELECT E.STAFF_ID, E.FNAME, E.LNAME, B.LOCATION

FROM EMPLOYEE E

INNER JOIN BRANCH B ON E.BRANCH_ID = B.BRANCH_ID

WHERE B.BRANCH_ID = 'a123';

STAFF_ID	FNAME	LNAME	LOCATION
E6	Sarah	Wilson	riffa
E7	Matthew	Martinez	riffa
E8	Olivia	Garcia	riffa
E9	Daniel	Lopez	riffa
E10	Sophia	Perez	riffa

5 rows returned in 0.01 seconds [Download](#)

10) retrieve customer information for those who booked a Suite room, along with the country from the HOTEL table:

SELECT C.CUSTOMER_ID, C.FNAME, C.LNAME, H.COUNTRY

FROM CUSTOMER C

JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID

JOIN HOTEL H ON B.HOTEL_ID = H.HOTEL_ID

WHERE H.ROOM_TYPE = 'Suite';

CUSTOMER_ID	FNAME	LNAME	COUNTRY
C003	Michael	Johnson	London
C004	Emily	Davis	Germany

2 rows returned in 0.02 seconds [Download](#)

11) displays customer names with passport number and flight details:

```
SELECT C.FNAME, C.LNAME, C.PASSPORT, F.AIRLINE, F.COUNTRY, F.CLASS_TYPE, F.PRICE
```

```
FROM CUSTOMER C
```

```
JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID
```

```
JOIN FLIGHT F ON B.FLIGHT_ID = F.FLIGHT_ID;
```

FNAME	LNAME	PASSPORT	AIRLINE	COUNTRY	CLASS_TYPE	PRICE
Michael	Johnson	3456789012	Gulfair	LONDON	First Class	1200
John	Doe	1234567890	Gulfair	GERMANY	Economy	700
Emily	Davis	4567890123	Gulfair	GERMANY	First Class	1500
John	Doe	1234567890	Gulfair	JORDAN	Economy	520
Jane	Smith	2345678901	Gulfair	JORDAN	Business	820

5 rows returned in 0.01 seconds [Download](#)

12) calculate the sum of the total cost of booking with the car rental cost :

```
SELECT C.BOOKING_ID,
```

```
(CR.PRICE_PER_DAY * CR.NUM_OF_DAY) AS CAR_RENTAL_COST,
```

```
C.TOTAL_COST + (CR.PRICE_PER_DAY * CR.NUM_OF_DAY) AS UPDATED_TOTAL_COST
```

```
FROM BOOKING C
```

```
JOIN CARRENTAL CR ON C.BOOKING_ID = CR.BOOKING_ID;
```

BOOKING_ID	CAR_RENTAL_COST	UPDATED_TOTAL_COST
B003	150	1510
B005	120	2380

2 rows returned in 0.01 seconds [Download](#)

13) retrieve the employees working in branches with a postcode of 4567:

```
SELECT E.STAFF_ID, E.FNAME, E.LNAME, B.POSTCODE
```

```
FROM EMPLOYEE E
```

JOIN BRANCH B ON E.BRANCH_ID = B.BRANCH_ID

WHERE B.POSTCODE = '4567';

STAFF_ID	FNAME	LNAME	POSTCODE
E1	John	Doe	4567
E2	Jane	Smith	4567
E3	Michael	Johnson	4567
E4	Emily	Brown	4567
E5	David	Lee	4567

5 rows returned in 0.00 seconds [Download](#)

14) display customer information who choosed the first class and their flights details :

SELECT C.CUSTOMER_ID, C.FNAME, C.LNAME, F.AIRLINE, F.COUNTRY, F.CLASS_TYPE, F.PRICE

FROM CUSTOMER C

JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID

JOIN FLIGHT F ON B.FLIGHT_ID = F.FLIGHT_ID

WHERE F.CLASS_TYPE = 'First Class';

CUSTOMER_ID	FNAME	LNAME	AIRLINE	COUNTRY	CLASS_TYPE	PRICE
C003	Michael	Johnson	Gulfair	LONDON	First Class	1200
C004	Emily	Davis	Gulfair	GERMANY	First Class	1500

2 rows returned in 0.01 seconds [Download](#)

15) show the names, phone numbers, and emails of employees working in the Riffa branch:

SELECT E.FNAME, E.LNAME, E.STAFF_ID, E.PHONE_NUMBER, E.EMAIL

FROM EMPLOYEE E

JOIN BRANCH B ON E.BRANCH_ID = B.BRANCH_ID

WHERE B.LOCATION = 'riffa';

FNAME	LNAME	STAFF_ID	PHONE_NUMBER	EMAIL
Sarah	Wilson	E6	67890123	sarah.wilson@example.com
Matthew	Martinez	E7	78901234	matthew.martinez@example.com
Olivia	Garcia	E8	89012345	olivia.garcia@example.com
Daniel	Lopez	E9	90123456	daniel.lopez@example.com
Sophia	Perez	E10	12345678	sophia.perez@example.com

5 rows returned in 0.01 seconds [Download](#)

16) calculate the sum of total cost in the BOOKING table for the Silver package:

```
SELECT SUM(TOTAL_COST) AS TOTAL_COST_SILVER_PACKAGE
```

```
FROM BOOKING
```

```
JOIN PACKAGE P ON BOOKING.PACKAGE_ID = P.PACKAGE_ID
```

```
WHERE P.PACKAGE_TYPE = 'silver';
```

TOTAL_COST_SILVER_PACKAGE
2470

1 rows returned in 0.01 seconds [Download](#)

17) count total number of booking per package :

```
SELECT PACKAGE_ID, COUNT(*) AS TOTAL_BOOKINGS
```

```
FROM BOOKING
```

```
GROUP BY PACKAGE_ID;
```

PACKAGE_ID	TOTAL_BOOKINGS
a100	2
c300	2
b200	1

3 rows returned in 0.00 seconds [Download](#)

18) count the total cost for each customer bookings:

```
SELECT CUSTOMER_ID, SUM(TOTAL_COST) AS TOTAL_BOOKING_COST FROM BOOKING GROUP
```

```
BY CUSTOMER_ID;
```

CUSTOMER_ID	TOTAL_BOOKING_COST
C002	1360
C001	2470
C003	2100
C004	2260

4 rows returned in 0.00 seconds [Download](#)

19) retrieve hotel details with "Four Season" in their name, along with their package types:

```
SELECT H.*, P.PACKAGE_TYPE
```

```
FROM HOTEL H
```

```
JOIN PACKAGE P ON H.PACKAGE_ID = P.PACKAGE_ID
```

```
WHERE H.HOTEL_NAME LIKE '%Four Season%';
```

HOTEL_ID	HOTEL_NAME	LOCATION	PACKAGE_ID	NUM_OF_STAR	ROOM_TYPE	PRICE_PER_NIGHT	COUNTRY	PACKAGE_TYPE
LON1	Four Season	London	a100	4	Standard	150	London	silver
PAR3	Four Season	5688 DC9	c300	5	Suite	350	Paris	premium

2 rows returned in 0.01 seconds [Download](#)

20)edit jhon doe email from john.doe@example.com to john.doe2@example.com:

```
UPDATE CUSTOMER
```

```
SET EMAIL = 'john.doe2@example.com'
```

```
WHERE CUSTOMER_ID = 'C001';
```

```
1 row(s) updated.

0.01 seconds
```

Group Members Contribution

It must show your meetings and contribution of each member (weekly meetings are recommended)

Manar: Writing || Dana: Designer || Aljazi: Planner || Noora: Leader

Session 1: Rever, Bahrain Bay 6 Nov 2023 10 am – 1:30 pm Writing of introduction, background, problem, and user requirements.

Session 2: Black Drop Café 7 Nov 2023 5 pm – 7 pm Discussion of tables and attributes.

Session 3: Gra Café 8 Nov 2023 11 am – 1:30 pm Implementation of attributes in SQL.

Session 4: Chocomelt 13 Nov 2023 11 am – 1 pm Implementation of data elements.

Session 5: Google Meet 17 Dec 2023 11:30 am - 2 pm Completed SQL implementation + SQL queries.

Session 6: Uncle Sam Café 20 Dec 2023 11 am – 2:30 pm SQL queries + Discussion of ER, EER & connection traps.

Session 7: Teams Meeting 21 Dec 2023 3 pm – 4 pm Discussion with the Doctor about ER.

Session 8: Beand Coffee Bar 23 Dec 2023 10 am – 1:30 pm Updating and modifying the ER and EER model.

Session 9: The Collective Hub 24 Dec 2023 10 am – 5:30 pm Drawing ER, EER, connection trap diagram & SQL.

Session 10: Google Meet 25 Dec 2023 9 pm – 3 am Finalizing the Data Element and SQL queries.

Session 11: Google Meet 26 Dec 2023 10:30 pm – 12:30 am Normalization.

Session 12: Google Meet 27 Dec 2023 pm – pm presentation practice.

Conclusion

In conclusion, we managed to create a full database management system for our travel agency that included elements like bookings, transportation rentals, flight details, and more. With the use of this system, all aspects are organized for our customers to have a smooth and easy trip excluding any difficulties or obstacles.